

Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC)

BSC (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

• GPA - 3.94/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

Research Interests

My research interests include **neuro-symbolic robotic manipulation**, **geometry-aware decision-making**, **long-horizon planning**, and **foundation model-based robot learning**. I am interested in combining **neuro-symbolic planning**, **Task and Motion Planning (TAMP)**, and **Vision-Language-Action (VLA) models** to build robots that can reason, plan, and act under **uncertainty** in complex **human-centered tasks**, such as **table-top Mahjong playing**.

Experience

Beijing Academy of Artificial Intelligence (BAAI)

RESEARCH ASSIST, ADVISED BY [SEN CUI](#)

• **Research Topics:** World models

Beijing, China

August 2026- Now

Tsinghua University (THU)

RESEARCH ASSIST, ADVISED BY [PROF. MENGDI XU](#)

• **Research Topics:** Manipulation, Neuro-Symbolic planning, VLA models

Beijing, China

July 2026- Now

Shanghai Jiao Tong University (SJTU)

RESEARCH ASSIST AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

• **Research Topics:** optimal control, generative model, grasping

Robot arm: Project leader

D.I.V.E FORUM: Organizer

Underwater 3D reconstruction: Project leader

WBCD 2nd dual arm robot: member

Shanghai, China

June 2025 - June 2026

The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C.H. NGAI](#)

• **Research Topics:** federated learning, fairness, spatio-temporal graphs.

Hong Kong SAR, China

March 2025 – August 2025

Publications

PREPRINTS

[P1] F²STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting

Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai

Under review, 2025

CONFERENCE PROCEEDINGS

[C4] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai

AAAI, 2026

[C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration

Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg, Wei Pu

IROS, 2025 (Oral) [\[Paper\]](#)

[C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding

Mingxiao Huo*, [Jiayi Zhang](#)*, Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen

ICML Workshop, 2025 [\[Paper\]](#)

[C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation

Yijie Li*, Hewei Wang*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev

ICLR Workshop, 2025 [\[Paper\]](#)

JOURNAL ARTICLES

[J2] **TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization**
Jiayi Zhang*, Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev
arXiv preprint, 2025 [Paper]

[J1] **RAINER: A Robust Ensemble Learning Grid Search-Tuned Framework for Rainfall Patterns Prediction**
Zhenqi Li*, Junhao Zhong*, Hewei Wang, Jinfeng Xu, Yijie Li, Jinjiang You, Jiayi Zhang, Runzhi Wu, Soumyabrata Dev
arXiv preprint, 2025 [Paper]

Honors & Awards

2026	Outstanding student (Top 1% student) , UNNC Award for Outstanding Students (UG)	Ningbo China
2025	Provost's Scholarship (Top 5% student) , University Academic Excellence Scholarship	Ningbo China
2025	Third Prize & Best Researcher Award , 9th Golden Interns Award	Ningbo China
2025	Honorable Mention , Interdisciplinary Contest in Modeling 2025	Online

Academic Services

Reviewer NeurIPS, AAAI, AISTATS 2025–Present

Leadership/Mentorship/Social

Semester Peer Mentor Ningbo China
UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC) Sep. 2025–Jun. 2026
• Provided academic guidance and practical resources for freshmen.

Skills

Programming Python, C++, $\text{\LaTeX}$, MATLAB, HTML, CSS, R
Frameworks PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS
Models CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion
Platforms Unitree Z1 Pro, Franka Research 3

Additional

Language Chinese (native), English (working proficiency)
Interests Running, hiking, mahjong, chessing, pickleball, roguelike and strategy games